

CC7.2 Optimization

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Summary:

This program implements a simply library search engine that users can use to search and sort books.

Time complexity by class:

Book class

- Consists of getters and setters all of which run at $O(1)$.

Library

- The getters and setters ('setBooks' and 'getBooks') run at $O(1)$.
- For 'searchBookByKeyword', I used the String contains method which, upon research, has some sources say it runs at $O(n)$ and others run at $O(nm)$. With that being applied in the for loop, the time complexity could range between $O(n^2)$ to $O(n^3)$ depending on the size of m .
- For 'viewAllBooks', it runs in $O(n)$ as it's a simple for loop to display all books in the list
- For 'loadBooks', it has a for loop that loops through the entire file and for each line, the split method is called which has a time complexity of $O(n)$. That makes this method's worst time, $O(n^2)$

LibraryMenu

- For 'LibraryMenu', the time complexity depends on whether there is a saved serialization of the library. If so, it calls 'setBooks' which runs at $O(1)$. If not, it calls 'loadBooks', which takes about $O(n^2)$.
- For 'sortMenu' and 'mainMenu', it runs at $O(1)$. Although there's a while loop, it's not actually looping through data but rather loops until the user provides a valid number
- For 'displayMenu', its runtime depends on what the user decides to do as it's main function is to call other functions. Without the functions, the method runs at $O(1)$ meaning that any choices the user makes in this method would take $O(1) * \text{time complexity of the function that is called to execute the action}$.

LibrarySerialization

- For both 'saveLibrary' and 'loadLibrary', writeObject and readObject is called. I couldn't find any specific information about these two's time complexity or detailed implementation of each method so I will assume both runs at $O(n)$ where n is the number of attributes an object has that needs to be written. For 'loadLibrary' specifically, It can be $O(n)$ if the file exists or $O(1)$ if the file doesn't exist or is empty.

Main

- The time complexity of the main function depends on the user's choices after executing

- The time complexity of the main function depends on the user's choices after executing the 'displayMenu' method. Otherwise, the rest of the code has a time complexity of $O(1)$.

SortUtil

- bubbleSort, insertionSort, and quickSort all have a worst case of $O(n^2)$.

UserInteractionLogger

- All methods in this class have a worst case of $O(1)$. 'logSearch', 'logSort', 'logViewAllBooks' all call the same method 'log'. The 'log' method simply opens the file, appends the string, and closes the file. No looping through data is done.

Optimization:

- Modified 'searchBookByKeyword' method to return a list of books that contain the keyword instead of the first book in the list

Optimization suggestions:

- Implementing methods to add and remove books from the library
- Implement methods to find lists of books by title, author, or publication date